

ROVANIEMI RAUTATIEASEMA, Finland

WMO: 028470

Lat: 66.498N

Lon: 25.708E

Elev: 82

StdP: 100.34

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.1	-25.1	-31.2	0.2	-28.0	-27.8	0.3	-25.0	8.0	-2.6	7.1	-2.5	0.7	60	0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	26.3	17.0	24.2	16.1	22.3	15.3	18.8	23.2	17.5	22.3	16.4	20.6	2.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.4	20.9	15.8	11.4	18.8	14.7	10.5	18.0	53.7	22.9	49.6	22.4	46.3	20.4	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.5	5.7	5.1	-30.4	27.8	3.8	2.5	-33.2	29.6	-35.5	31.0	-37.6	32.4	-40.4	34.2
			-30.4	20.4	3.6	1.6	-33.0	21.5	-35.2	22.4	-37.2	23.3	-39.9	24.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.2	-11.7	-9.8	-5.1	1.1	8.2	12.8	16.7	14.0	9.1	1.7	-3.4	-7.4
	DBStd	10.61	6.65	7.56	4.95	3.70	4.60	3.91	3.55	3.37	3.33	3.83	5.15	6.51
	HDD10.0	3329	672	553	469	267	92	16	1	6	54	257	401	540
	HDD18.3	5905	931	787	727	517	317	172	75	140	276	516	651	798
	CDD10.0	501	0	0	0	1	34	99	207	130	29	0	0	0
	CDD18.3	34	0	0	0	0	1	5	23	6	0	0	0	0
	CDH23.3	275	0	0	0	0	21	36	181	38	0	0	0	0
	CDH26.7	49	0	0	0	0	2	4	38	6	0	0	0	0
Wind	WSAvg	2.3	1.9	2.2	2.4	2.3	2.7	2.6	2.3	2.3	2.2	2.2	2.2	2.2
Precipitation	PrecAvg	608	43	36	33	33	44	62	77	65	59	54	54	48
	PrecMax	822	72	74	68	83	99	108	131	149	118	91	94	97
	PrecMin	428	16	4	10	10	8	13	20	9	12	14	14	14
	PrecStd	108	16	18	15	17	20	27	30	35	28	21	21	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.4	4.0	6.5	16.6	25.9	27.3	30.2	27.5	19.1	11.0	6.1	3.1
		MCWB	0.8	2.3	2.6	8.8	14.2	17.5	19.6	19.7	13.8	9.0	5.7	2.4
	2%	DB	0.1	1.8	4.3	11.3	22.9	24.2	27.5	23.8	17.3	9.5	4.8	1.8
		MCWB	-0.1	0.9	1.3	5.5	14.0	15.7	17.4	16.3	12.7	7.8	4.3	1.2
	5%	DB	-0.9	0.8	2.7	8.8	19.2	21.8	25.2	21.6	15.9	8.3	3.6	1.1
		MCWB	-1.1	0.3	0.5	4.2	11.6	13.9	16.7	15.6	12.3	7.1	3.2	0.7
	10%	DB	-2.3	-0.2	1.4	6.9	16.3	19.6	23.3	19.6	14.3	7.1	2.4	0.4
		MCWB	-2.6	-0.7	0.0	3.2	9.9	12.8	16.1	14.8	11.8	6.1	2.1	0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.9	2.4	3.1	8.9	15.4	18.7	20.9	20.0	15.3	10.0	5.7	2.6
		MCDB	1.2	3.6	5.6	16.1	22.7	24.6	25.3	25.0	17.4	10.5	5.9	3.1
	2%	WB	0.0	1.2	1.7	6.4	13.9	17.1	19.4	18.0	13.8	8.1	4.2	1.4
		MCDB	0.1	1.6	3.4	10.2	21.1	23.0	23.4	22.1	15.9	8.9	4.5	1.7
	5%	WB	-1.1	0.3	1.0	4.7	12.5	15.7	18.2	16.6	13.0	7.2	3.3	0.8
		MCDB	-1.0	0.8	2.0	8.1	18.7	20.0	23.0	19.7	15.0	8.3	3.5	1.0
	10%	WB	-2.7	-0.7	0.3	3.5	10.6	14.0	17.0	15.5	12.1	6.2	2.2	0.1
		MCDB	-2.3	-0.2	1.3	6.3	15.2	17.3	21.7	18.2	13.8	7.1	2.4	0.4
Mean Daily Temperature Range	5% DB	MDBR	7.0	7.0	8.9	8.5	9.1	8.2	8.6	7.9	6.8	5.0	4.7	6.0
		MCDBR	7.7	5.5	8.3	11.7	14.1	11.4	11.7	11.1	9.4	6.4	4.4	4.6
		MCWBR	7.8	4.9	6.1	6.8	6.4	4.9	4.3	4.8	5.4	5.0	4.2	4.5
	5% WB	MCDBR	7.9	5.3	6.0	10.4	12.8	9.2	9.9	8.7	7.1	5.6	4.2	4.7
		MCWBR	8.0	4.7	4.8	6.4	6.4	5.0	4.5	4.6	4.6	4.8	4.1	4.7
Clear-Sky Solar Irradiance	taub	0.199	0.267	0.299	0.334	0.345	0.346	0.360	0.343	0.319	0.291	0.347	N/A	
	taud	1.935	2.103	2.166	2.242	2.445	2.469	2.448	2.479	2.532	2.346	2.640	N/A	
	Ebn at Noon	203	569	749	821	857	863	836	814	754	559	214	N/A	
	Edh at Noon	16	54	85	103	94	94	94	82	62	43	16	N/A	
All-Sky Solar Radiation	RadAvg	0.07	0.55	1.97	3.63	4.71	5.19	4.88	3.54	1.87	0.65	0.13	0.01	
	RadStd	0.01	0.10	0.21	0.32	0.43	0.40	0.46	0.32	0.20	0.13	0.02	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (42 neighbors)	+0.55	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page